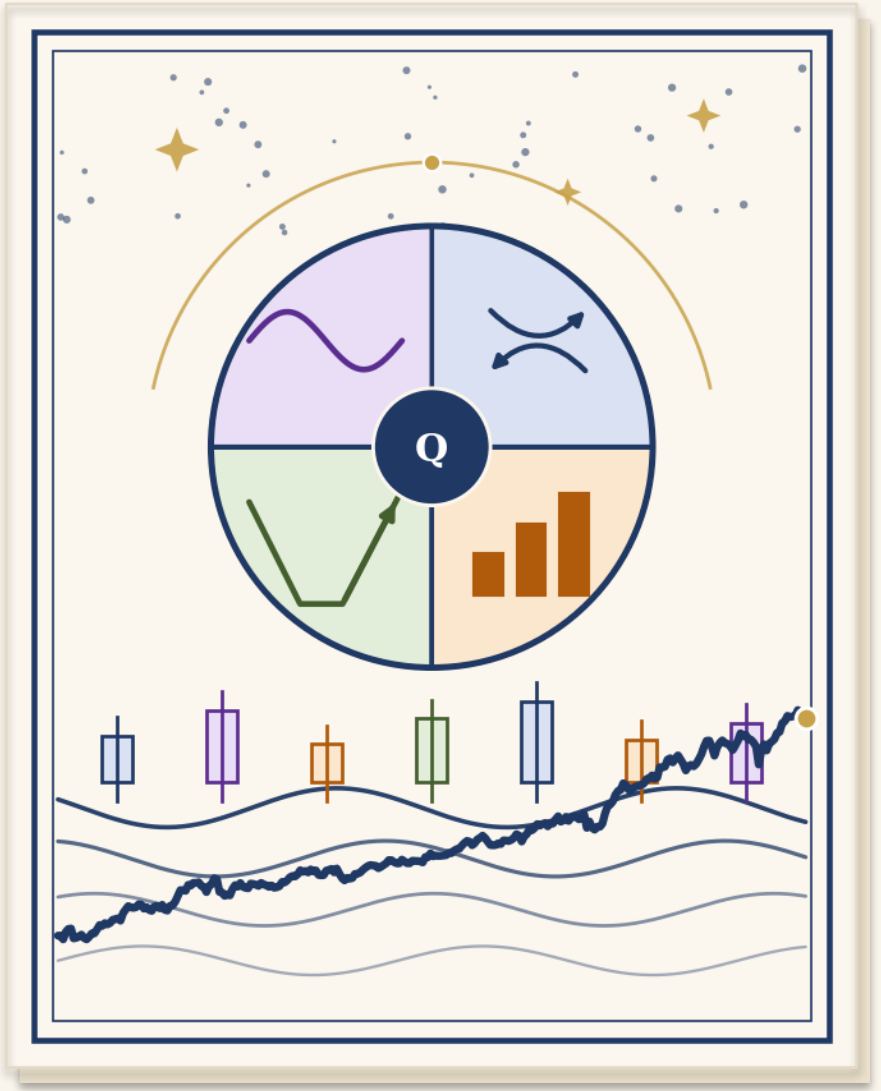
THE

Quadrant Strategy.



Four Market Structures

One book, four engines — each harvesting a different named premium.

1.54 Sharpe · **15.4%** CAGR · **−8.1%** Max DD · **+14.1%** in 2022



The book, animated

Equity curve, Apr 2021 — Oct 2026, net of costs. **+121.0% total**

Andrew Wiraatmaja · Edward Sujono · Michael Japardi · Victoria Chan · Wielly Halim

SPEAKER 1 · ANDREW

Tariffs, a yen unwind, a crypto winter

Five regimes in five years. Each paid a different premium and punished a different flaw. Here is which sleeve answered each one.

2021–22 · Crypto winter	BTC fell 74% peak-to-trough; buy-and-hold wore all of it.	Wheel: max drawdown −43%, and USD 476K of modelled premium across the window.
2022 · The rate shock	Stocks and bonds fell together; the classic hedge failed.	IDX carried the year: coal and nickel +98.7% on the energy shock. E1 −2.2%, the wheel bled, dip sat in cash (−1.1%). Book: +14.1%.
Aug 2024 · The yen unwind	A three-day global deleveraging as the carry trade burst.	Book: +0.4%. Nothing in the book trades the yen; nothing had to.
Apr 2025 · The Trump tariff shock	Policy whiplash: tariffs announced, paused, re-announced within weeks.	The dip sleeve bought the tariff dip (+9.3% in a week). Book: +1.1%.
2023–26 · The gold supercycle	Speculative positioning pinned at the 79th percentile (Oct 2026).	IDX loads on gold (β +0.38); the gold basket and COT brake are design variants outside the book.

No single strategy survived all five regimes. The book is built so no single headline can fire every sleeve at once.

Why a book, not a strategy

Sleeve	Premium	Market	What kills it
1. Mean reversion	Relative value	SGX + US equities	Crowding; regime change
2. BTC wheel	Volatility	Bitcoin options	A full bear market, held naked
3. IDX commodities	Commodity beta	Indonesian equities	Commodity downcycles; IDR
4. Dip buying	Short-term reversal	Nasdaq-100 equities	A bear that keeps dipping; no stop loss

Measured on the common window (Apr 2021 to Oct 2026), every pair of sleeves correlates between **−0.09 and +0.23**. The four signals are differentiated, not independent — the wheel carries BTC beta and IDX loads on gold — so the book can be more than the sum of its parts, if it is sized honestly.

The shared flaw, stated before the results: sleeves 2, 3 and 4 are all long risk. What stands against a crash is Sleeve 1's near-zero measured factor exposure (R^2 0.01), Sleeve 4's trend filter (it held −1.1% in 2022 by sitting mostly in cash), and the book brake. The portfolio slide tests whether that is enough.

1.54
Book Sharpe (common window)

≤ 0.23
Highest pairwise sleeve correlation

+14.1%
Book return in 2022

−8.1%
Book max drawdown

Mean reversion engine (PCA statistical arbitrage)

0.67
Sharpe, net (2019–2026)

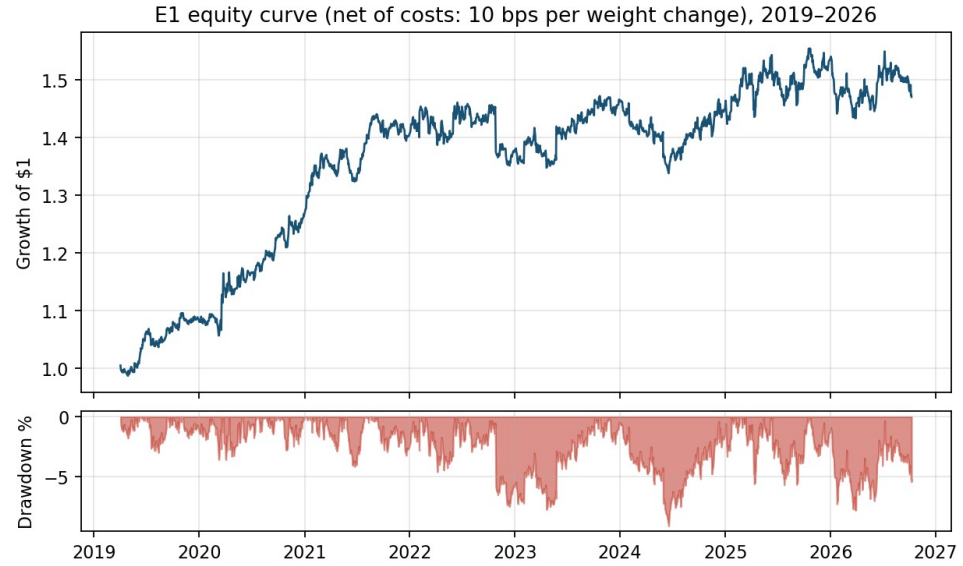
−9.1%
Max drawdown

1,541
Trades · 59.1% win rate

0.06
Sharpe since 2024 — decaying

Frozen spec. Residuals from a 3-factor PCA model (60-day window) on 34 names (9 SGX + 25 US). Trade the OU s-score: open at ±1.5, close at ∓0.5, stop at $|s| > 3.5$, half-life ≤ 30 days. Costs: 10 bps on every change in position weight (entries, exits and daily rebalancing).

Net results, Apr 2019 to Oct 2026	
Return	5.6%/yr
Sharpe	0.67
Volatility	8.5%
Max drawdown	−9.1%
Trades / win rate	1,541 / 59.1%
Avg hold / profit per trade	11.8 days / +2.7 bps



Equity curve and drawdown, net of costs. The engine made money through the 2020 crash; 2022 was a small loss (−2.2%) once every rebalance is charged, and its flat years are recent.

Lecture exhibits: factor attribution (A6) · cointegration and residual diagnostics (A9)

The flaw, on the record: the edge is real and decaying, so this engine is the book's ballast at 40%, not its return promise.

Bitcoin options wheel

0.65 vs 0.38
Sharpe vs buy-and-hold

$476K
Modelled gross premium, 289 sales

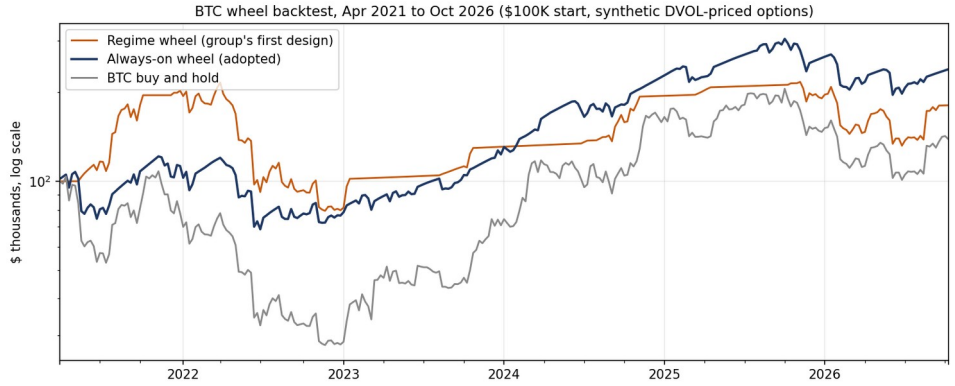
−43% vs −74%
Max DD vs buy-and-hold

≈22%
1-wk 99% ES — the tail (A7)

Mechanism. Weekly cycle, 25-delta options, 1 week to expiry, priced off Deribit's DVOL index. Flat: sell cash-secured puts. Holding: sell covered calls. Cash earns the T-bill rate. Apr 2021 to Oct 2026, USD 100K.

	Sharpe	CAGR	Max DD
Regime wheel (first design)	0.48	11.3%	−63.3%
Always-on wheel (adopted)	**0.65**	**17.2%**	**−43.4%**
BTC buy and hold	0.38	6.2%	−74.3%

What the test killed: the first design sold puts only in bear regimes and calls only in bull regimes. In 2022 it returned −60.7%, close to buy-and-hold (−64%): once assigned, it sold nothing and rode the bear naked. The plain wheel kept harvesting calls through the decline and lost −32.0%.



Adopted vs rejected vs buy-and-hold. The regime overlay is kept in the rejection log (A3), not buried.

Lecture exhibits: implied vs GARCH volatility (A8) · tail risk, EVT (A7)

Always-on, never idle: the regime filter was the flaw. Short volatility by construction, so it stays the smallest sleeve at 15%. Its 17.2% CAGR is the wheel's own window from 2 Apr 2021; on the book's common window it is 16.7%.

IDX commodity sleeve (weekly signals)

1.11 vs 0.25
Sharpe vs IHSG benchmark

30.2%
CAGR (IDR)

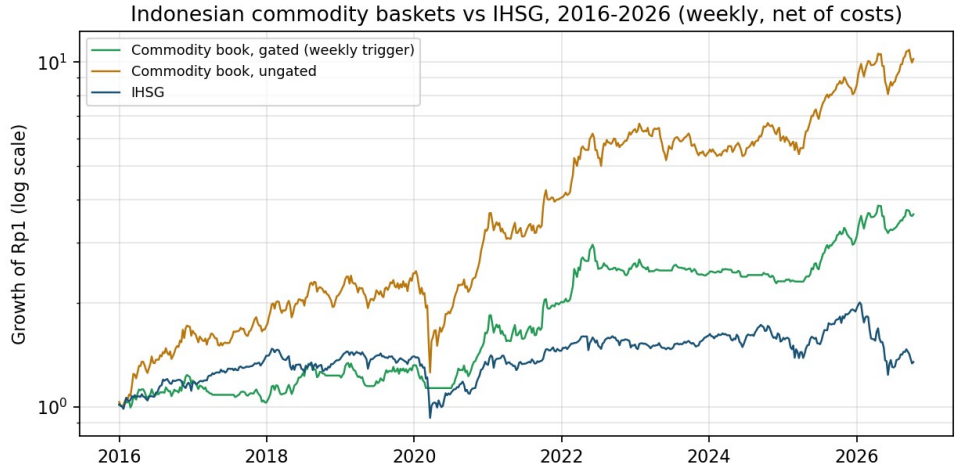
+0.38
IDX beta to gold (A6)

OFF · OFF
Design-variant gates, 9 Oct

Design. Equal-weight Indonesian commodity baskets, weekly, 25 bps per side, IDR. Coal: ADRO, PTBA, ITMG, BYAN. Nickel: ANTM, INCO, MDKA, NCKL, MBMA. Gold miners: ANTM, BRMS, PSAB, MDKA, ARCI.

	Sharpe	CAGR	Max DD
Coal + nickel (growth)	1.11	30.2%	−41.8%
IHSG benchmark	0.25	4.1%	−38.4%

Findings: (1) the alpha is **asset selection**: IHSG is bank-heavy and dilutes the commodity torque. (2) Trend gates are **insurance, not alpha** (coal drawdown −51% → −22%, no Sharpe added). (3) In the gated variant, a COT positioning brake lifted the gold-miner basket's Sharpe 0.81 → 0.86; the naive version failed (A3).



Weekly books vs IHSG, 2016 to 2026. Returns in IDR; SGD investors earn less after FX. The book series is the ungated coal + nickel growth baskets; gates and the gold basket are design variants.

CAGR here is the annualised mean weekly return; geometric: IDX 30.3%, IHSG 2.8%. · Exhibits: A6 · A9

Selection is the alpha: concentrated commodity torque the benchmark dilutes, with gates that insure rather than pretend to predict.

Dip buying, Nasdaq-100

1.07
Sharpe, long run 2015–2026

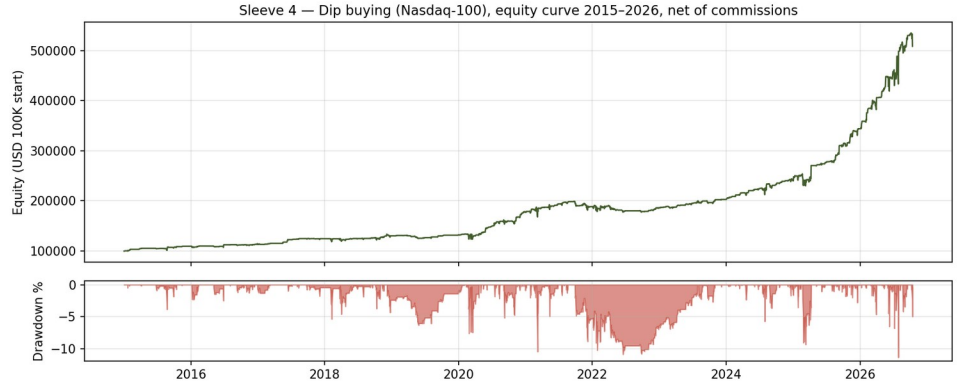
1.70
Sharpe since 2024

−1.1%
2022: sat the bear out

≈13%
Average capital use

Design. Our rebuild of a published dip-buying rule set, verified against its reported results. Nasdaq-100 names above their 200-day average; a one-day drop over 3% at high ATR triggers a next-day limit buy 0.9×ATR below the close. Exits: a +0.5×ATR target, a close above the prior day's high, or day 11. Max 10 positions. No stop loss.

Long run 2015–2026	Sharpe 1.07 daily (1.50 weekly), CAGR 14.8%
Trades / win rate / hold	**853 / 69.9% / 4.7 days**
Max drawdown	−11.3%
COVID crash, Feb–Mar 2020	−3.7%
Since 2024 (published test window)	Sharpe 1.70, CAGR 39.1% (to 14 Oct 2026)



Equity curve and drawdown, 2015 to 2026, net of IBKR commissions. The flat stretches are the design working: no qualifying dips, no trades. Universe: today's constituents applied backwards (slide 10).

Lecture exhibits: factor attribution (A6) · tail risk, EVT (A7)

A long-only sleeve that survives bears by absence: the 200-day filter keeps it in cash when dips turn into downtrends. Its risk is the mirror image: a crash that starts at all-time highs.

The whole falls less than every part

1.54
Book Sharpe (common window)

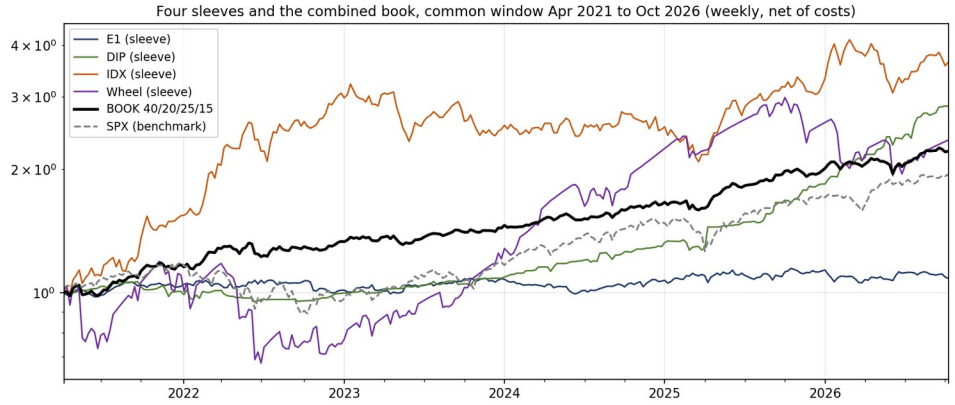
+14.1%
2022: BTC −64%, equities −19%

−8.1%
Book max drawdown

+0.07
Book equity beta (A6)

Allocation, quarterly rebalanced: **E1 40% Dip 20% IDX 25% Wheel 15%**. Sleeves are aggregated in native currencies (IDX in IDR, unhedged): a model composite, not a single-currency fund return.

	Sharpe*	CAGR	Vol	Max DD
E1	0.22	1.5%	8.2%	−8.4%
DIP	1.73	20.8%	11.3%	−10.9%
IDX	1.04	26.3%	25.5%	−35.3%
Wheel	0.65	16.7%	32.4%	−43.4%
BOOK	**1.54**	**15.4%**	**9.6%**	**−8.1%**
SPX	0.84	12.7%	15.7%	−24.8%



Correlations: E1–DIP −0.09, E1–IDX −0.03, E1–Wheel −0.01, DIP–IDX −0.08, DIP–Wheel +0.16, IDX–Wheel +0.23. The grey dashed line is the S&P 500: +94.3% over the window with a −24.8% drawdown, against the book's +121.0% and −8.1%. The dip sleeve trades E1's market and still correlates at −0.09: it deploys only ≈13% of its capital on average, with positions open on about half of days. Lecture exhibits: A5 · A7 · A6.

Stress record: 2022 +14.1%; Aug 2024 +0.4%; Apr 2025 +1.1%. The 12% vol-target overlay (Sharpe 1.45, DD −10.7%) levers the book up from 9.6% vol: CAGR 19.0%, deeper drawdown. Kept out.

*Sharpe = annualised mean weekly return ÷ standard deviation, no risk-free rate deducted (all rows); net of T-bills (avg 3.4%): book 1.18, SPX 0.63.

Diversification is still the strategy at book level: Sharpe 1.54, max drawdown −8.1%, positive in all three stress windows; 2022 is IDX-driven (+98.7%), without which the other sleeves lose about 7%. SPX: Sharpe 0.84, drawdown −24.8%, −19.4% in 2022.

SPEAKER 4 · MICHAEL

Operating plan

Discipline before performance.

- Rules frozen 1 Oct; the freeze log at right records two changes made inside the window, with reasons.
- Weekly tear sheet per sleeve and for the book, produced at every Friday close in the window.
- Book brake: halve gross at −5% drawdown; stop and review at −8% (backtests exclude the brake). October crossed −5% in the week ending 16 Oct; test returns are unbraked.

Reporting, in the brief's own metrics: annual return, Sharpe, maximum drawdown and drawdown count, win rate, volatility, number of trades, average holding period, profit per trade.

Reported per sleeve and for the book, with trade counts beside every ratio. A few weeks of live data can validate operations and discipline, not a premium, and we will not present it as if it does.

Freeze log: frozen 1 Oct; two changes inside the window, both on pre-window evidence: (1) 15 Oct, Sleeve 4 CTA futures replaced by dip buying (verified 1.07 vs 0.34, A3); (2) 22 Oct, E1 costing rerun at full turnover (returns fell). **Brake, quantified:** at or below −5% in 10 weeks (5 episodes); −8% breached once (5 Jun 2026, −8.12%); return effect not simulated. The swap was no response to the live loss: the dip sleeve fell

The operating rhythm

DAILY

E1 s-scores and dip triggers checked. Orders only from frozen rules; every fill logged.

WEEKLY

IDX coal and nickel baskets rebalanced; gate status logged. The wheel rolls the next 25-delta BTC contracts.

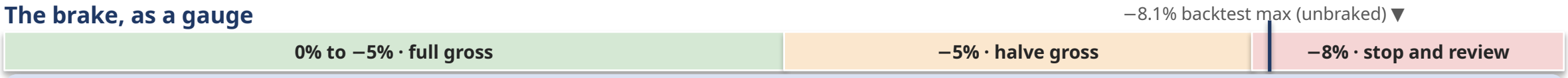
QUARTERLY

Book rebalanced back to 40 / 20 / 25 / 15. The weights are not negotiable in between.

FRIDAY CLOSE

Tear sheet per sleeve and for the book, in the brief's metrics, trade counts beside every ratio.

The brake, as a gauge



The window grades discipline, not luck: frozen rules, Friday tear sheets, and a brake whose −5% line the book crossed in the week ending 16 Oct.

Limitations, stated plainly

- **Wheel:** no option quotes were used. Premiums are Black–Scholes at one flat vol (Deribit's 30-day DVOL) on 1-week 25-delta options: no skew, no bid–ask, no term structure, less a 5% haircut. The $476K premium and the A8 vol gap are model outputs that likely flatter calm markets. Venue and assignment mechanics simplified.
- **E1:** the 34-name universe (9 SGX, 25 US) is today's large-cap list applied backwards; decay documented, not explained away; SGX borrow assumed at modelled cost.
- **Dip:** today's Nasdaq-100 applied backwards, so names that fell out after a decline are absent; a buy-the-dip rule is the design most flattered by this, so read its 1.73 Sharpe as an upper bound. Sizing is reconstructed. Idle cash earns nothing in the backtest.
- **IDX:** baskets are today's known winners (survivorship flatters levels; same-basket comparisons are insulated). The nickel basket held three names until Apr 2023 (NCKL and MBMA listed later), so 2022's +98.7% is a seven-stock result. IDR unhedged; coal and nickel are one cycle, not two.
- **Book:** the window starts Apr 2021 (BTC options history); no 2008/2020-style crash for the full book, and the live window is weeks long.

Where each sleeve is most exposed



Kill criteria, pre-committed

Sleeve	Kill rule
E1	Stop if live profit per trade is negative after 200 trades, or if the freeze log shows a parameter change.
Wheel	Stop if cumulative assignment losses exceed the trailing two years of premium collected.
IDX	Stop if the sleeve's live drawdown exceeds −42%, beyond its full-history backtest maximum (−41.8% since 2016; −35.3% on the common window).
Dip	Stop if sleeve drawdown exceeds −13%, beyond its backtest maximum of −11.3% (July 2026, long run and since 2024 alike): the design has no stop loss, so the drawdown limit is the stop.
Book	The −8% brake is the kill criterion for the combination itself.

The book brake, on one line

0% to −5%: full gross	−5%: halve gross	−8%: stop and review

Backtest maximum drawdown without the brake: −8.1% (Apr 2021 to Oct 2026). The −8% stop sits just inside it, so a live repeat is cut at the stop, not after it.

Four premia, four market structures, one book: sized by damage, frozen in advance, and reported the way we would want to read it.

CONCLUSION

What we built, and what would change our minds

1 · Four premia, no single dominant driver.

Pairwise correlations −0.09 to +0.23; the book's equity beta is +0.07; but three sleeves are long risk (slide 3). Diversification here is measured in sample, over 288 weeks.

2 · The book falls less than its parts.

Sharpe 1.54 vs 0.22 to 1.73 for the sleeves; maximum drawdown −8.1%, shallower than every sleeve; +14.1% in 2022. The dip sleeve rivals it, but deploys only ≈13% of capital on average (in cash on about half of days): it cannot be the whole book.

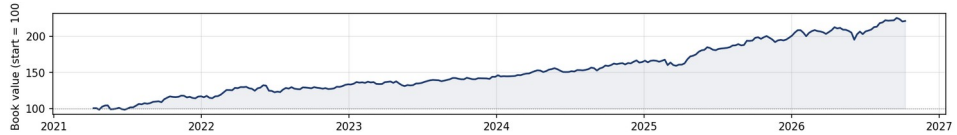
3 · Four designs dropped; the record is public.

Three designs died on their own evidence, and a fourth, the working CTA futures sleeve, was still substituted when the dip strategy verified stronger (A3).

4 · Sized by damage, frozen in advance.

EVT tails set the weights' conscience: the wheel is smallest because its shortfall is largest (A7). The brake is pre-committed (slide 11).

What would change our minds: a sleeve breaching its kill rule in the live window. We stop it, and we report that too.



The book: Apr 2021 to Oct 2026, 100 → 221.

THE HOOK, ANSWERED

Crypto winter	Wheel −43% vs −74%
Rate shock 2022	Book +14.1%
Yen unwind	Book +0.4%
Tariff shock	Book +1.1%
Gold supercycle	Indirect only: IDX gold β +0.38

The claim of this project is not that we found four edges. It is that we built a book that can survive being wrong about one of them.

Frozen specification sheet

Sleeve	Specification
E1	3 PCA factors, 60d window, s open ±1.5 / close ∓0.5, stop |s| > 3.5, OU half-life ≤ 30d; 34 names (9 SGX, 25 US); 10 bps on every change in position weight (entries, exits, daily rebalancing).
Wheel	Weekly Fri cycle; 25-delta 1-week options; puts when flat, calls when holding; full cash cover; T-bill on cash; 5% premium haircut; DVOL pricing (30d IV).
IDX	EW baskets, weekly; coal (ADRO, PTBA, ITMG, BYAN), nickel (ANTM, INCO, MDKA, NCKL, MBMA), gold (ANTM, BRMS, PSAB, MDKA, ARCI); sleeve gates: coal on BTU trend, nickel on own 40wk trend, gold on gold 40wk trend + COT Brake B; 25 bps/side; IDR. Book series: ungated coal + nickel growth baskets; gates and the gold basket are design variants (A3).
Dip	Nasdaq-100 (101 current constituents); close > 200d SMA; signal: 1-day drop > 3% and ATR(5)/close > 3%; next-day limit at signal close − 0.9×ATR(5); rank ATR(5)/close desc; max 10 positions; size = prior-close equity / 10 (reconstructed); exits: +0.5×ATR target, close > prior day high, or day 11; no stop; IBKR tiered commissions.
Book	40/20/25/15, quarterly rebalance; brake at −5% (halve) and −8% (stop) book drawdown.

FROZEN
Final specification, fixed Oct 2026; two changes logged (slide 9)

→

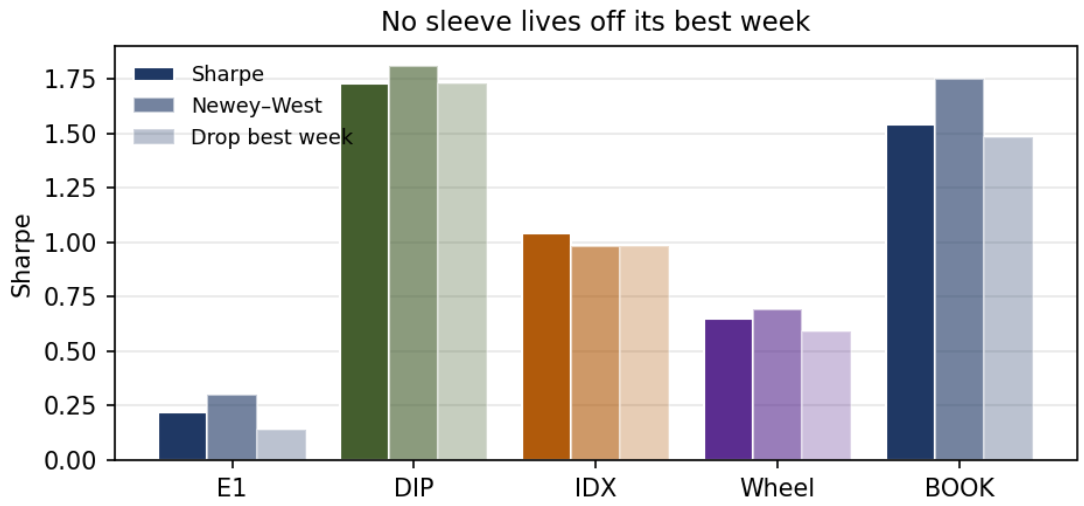
LIVE WINDOW
Oct 2026; final tear sheet at the 30 Oct close

→

NO SILENT CHANGES
Any change enters the freeze log, with reasons

Robustness (common window, weekly)

	Sharpe	Newey–West	Drop-best
E1	0.22	0.30	0.14
DIP	1.73	1.81	1.73
IDX	1.04	0.98	0.98
Wheel	0.65	0.69	0.59
BOOK	**1.54**	**1.75**	**1.48**
SPX	0.84	0.98	0.78



No sleeve, and not the book, depends on its single best week. Newey–West Sharpe is above raw Sharpe for every row except IDX, consistent with mild negative weekly autocorrelation in the others.

Parameter sensitivity: E1 Sharpe peaks at the frozen 3 factors (0.43/0.67/0.48/0.26 for 2/3/4/5 factors), so the frozen spec is the in-sample best on that axis. s-entry 1.75 gives 0.99 against the frozen 1.5 (drawdown −14.8% vs −9.1%); it was measured and deliberately not taken.

APPENDIX A3 · REJECTION LOG

What we tested and killed

Candidate	Evidence
CTA futures trend (built, then substituted)	Verified as designed (Sharpe 0.34; 2022 +17.4%), then replaced as Sleeve 4 by the dip strategy on the group's direction, for its stronger verified record (1.07 long run, 1.70 since 2024). Cost: the book no longer shorts bonds or the dollar.
BTC regime wheel	Sharpe 0.48 vs 0.65 always-on; 2022 −60.7% vs −32.0%. Replaced by the plain wheel.
COT Brake A (gold)	Exit at 90th positioning percentile: Sharpe 0.54 vs 0.56 (gold), 0.67 vs 0.81 (miners); out of the market 32% of weeks. Brake B (percentile + momentum) adopted instead.
CPO basket (IDX)	Sharpe 0.24 to 0.40 in every variant, drawdowns −53% to −68%. Dropped; gold miners took the third-basket slot.
Pairs trading	Set aside by the group as a well-trodden design before testing; the PCA engine generalises it.

5
Entries in this log

4
Killed or substituted

1
Set aside untested (pairs)

0
Killed designs still in the book

Data and anti-look-ahead protocols

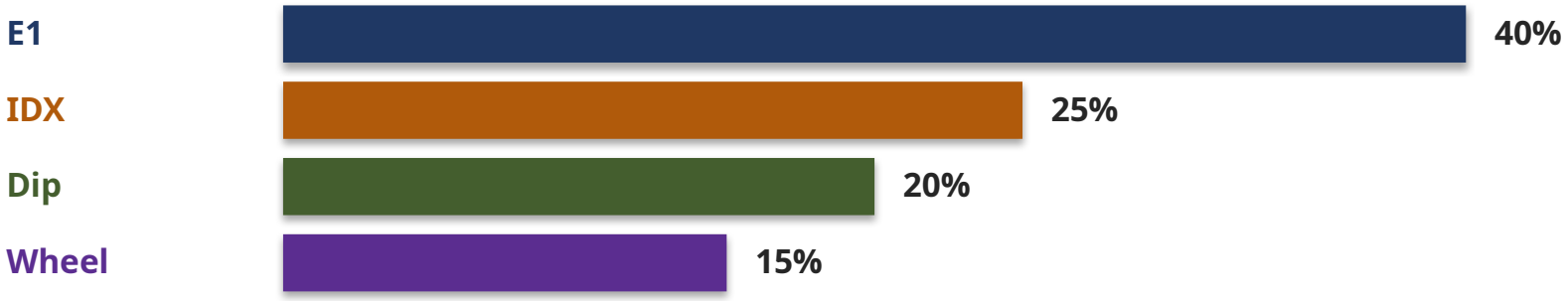
Sources. Prices: Yahoo Finance (equities, futures, BTC). Options: Deribit DVOL volatility index (daily, Mar 2021 onwards), options priced by Black–Scholes at 25 delta. Positioning: CFTC Commitments of Traders, COMEX gold, legacy noncommercial net as % of open interest. Rates: 13-week T-bill (^IRX).

Anti-look-ahead protocols. COT: a Tuesday report is used only at Friday evaluations at least 7 days later (it is published Friday, after the IDX close). All signals are computed on data dated on or before the decision close; execution is at the next session or the same close as stated per sleeve. Point-in-time caveat: IDX baskets and the E1 universe are today's constituents; the limitation is disclosed on slide 10, not buried here.



Why these weights (sensitivity)

Weights were set by design and frozen for the live window; the study validates the choice rather than fitting it. Anchor: risk parity on measured sleeve vols (E1 8.2%, Dip 11.3%, IDX 25.5%, Wheel 32.4%) gives ≈44/31/14/11, adjusted for E1's alpha decay, the dip sleeve's long-only equity exposure, and the wheel's short-vol tail.



Weights	Sharpe	CAGR	Vol	Max DD	2022
Chosen 40/20/25/15	**1.54**	**15.4%**	**9.6%**	**−8.1%**	**+14.1%**
Risk parity ≈44/31/14/11	1.78	13.8%	7.4%	−4.9%	+6.2%
Equal weight 25 each	1.48	18.4%	12.0%	−10.5%	+10.8%
E1-heavy 50/15/20/15	1.42	13.0%	8.9%	−7.4%	+9.8%

The optimum is no longer flat: parity reaches 1.78 by loading the low-volatility dip sleeve to 31%. The chosen weights accept 1.54 in exchange for the best crisis year (+14.1% in 2022 vs parity's +6.2%) and less riding on two equity sleeves. Either is defensible; the trade-off is disclosed, not hidden.

Factor attribution: the premia are different, measurably

Weekly return betas	SPX	BTC	Gold	Coal	R^2
E1	+0.03	−0.00	−0.02	−0.01	0.01
DIP	**+0.16**	+0.00	+0.08	+0.00	0.08
IDX	+0.03	**+0.08**	**+0.38**	**+0.07**	0.17
Wheel	+0.16	**+0.47**	−0.02	+0.02	0.66
BOOK	**+0.07**	**+0.09**	**+0.10**	**+0.02**	**0.41**

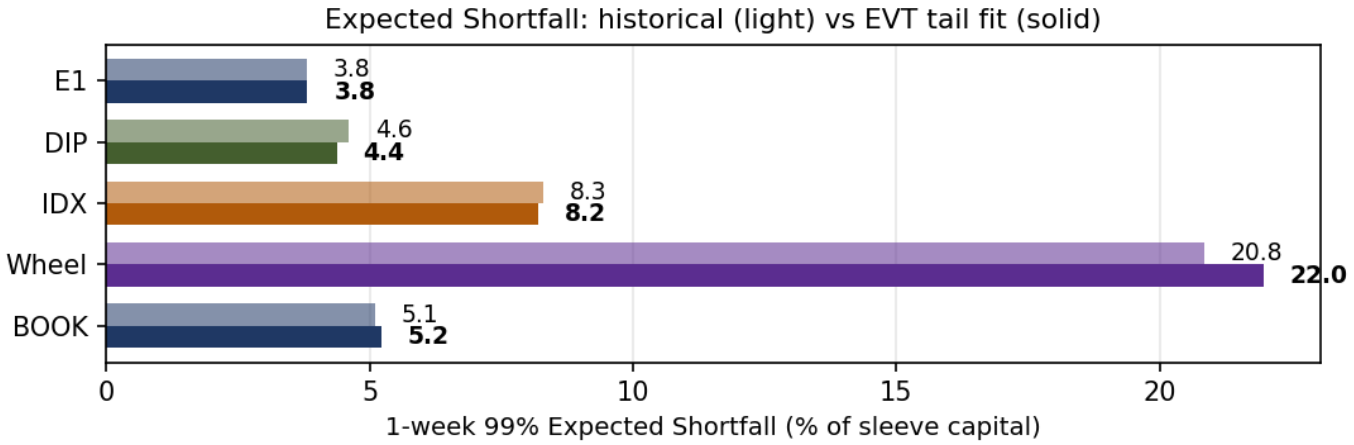
OLS of each sleeve's weekly returns on S&P 500, BTC, gold and coal futures, common window (Apr 2021 to Oct 2026, n = 288). **Bold: |t| > 2.6.** Cell shading deepens with |β|.

E1 loads on nothing (R^2 = 0.01): near-zero measured factor exposure, not neutrality by claim. The **book's equity beta is +0.07**: about +0.03 from the dip sleeve (+0.16 at a 20% weight), +0.02 from the wheel (+0.16 at 15%), the rest from E1 and IDX. The book's largest betas are to gold (+0.10) and BTC (+0.09). IDX loads on gold (+0.38) more than coal (+0.07): its nickel names (ANTM and MDKA are also gold miners) dominated the window.

Tail risk: one-week 99% VaR and Expected Shortfall

Losses, % of sleeve capital	VaR (hist)	ES (hist)	VaR (EVT)	ES (EVT)	Tail ξ
E1 (40%)	2.8	3.8	3.1	3.8	−0.03
DIP (20%)	4.1	4.6	3.6	4.4	−0.20
IDX (25%)	7.6	8.3	7.7	8.2	−0.53
Wheel (15%)	15.9	20.8	15.9	22.0	+0.12
BOOK	**4.5**	**5.1**	**3.5**	**5.2**	**+0.26**

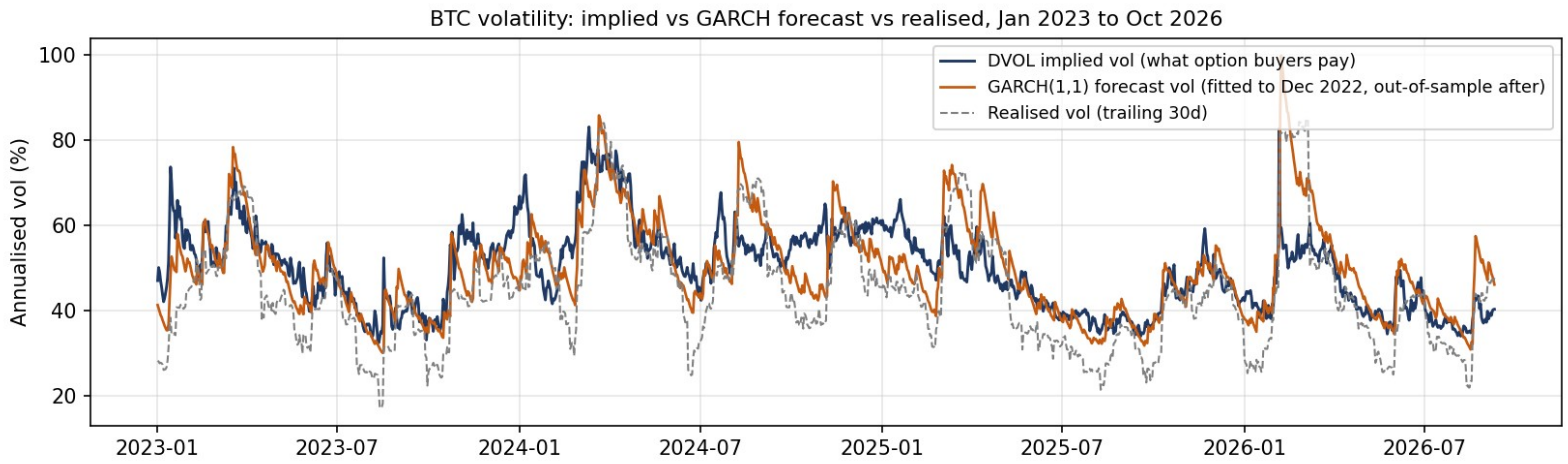
Historical quantiles on weekly returns plus an EVT peaks-over-threshold fit (GPD, 90% threshold, ≈29 exceedances; ξ is indicative, not precise), common window.



The **wheel owns the tail**, a 1-week ES of about 21 to 22% of its own capital, and is the smallest sleeve for exactly that reason.

The EVT fit and history now agree on the book (ES ≈ 5.1 to 5.2%): the dip sleeve's truncated tail (ξ −0.20; its exits cap the holding period, not the loss) thins the book's tail versus the earlier CTA-futures Sleeve 4 (A3). The −8% brake still sits beyond one very bad week.

The wheel's premium, measured: implied vs forecast vs realised vol



GARCH(1,1) on BTC daily returns, fitted Jan 2015 to Dec 2022: α 0.055, β 0.945, persistence ≈ 1.00. Volatility shocks barely decay, which is why a premium seller respects vol regimes.

Out-of-sample, Jan 2023 to Sep 2026 (means): DVOL implied 49.7%, GARCH forecast 49.7%, vol that subsequently arrived 44.4%.

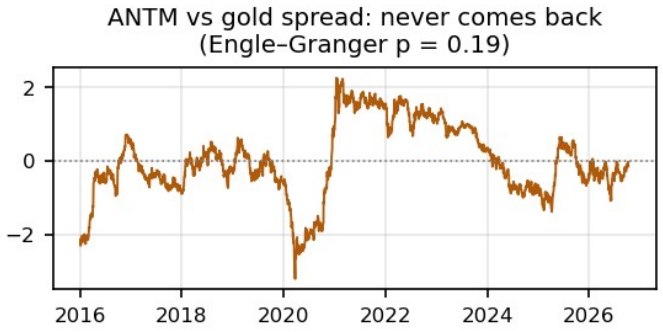
Implied exceeded subsequent realised vol on **69% of days**.

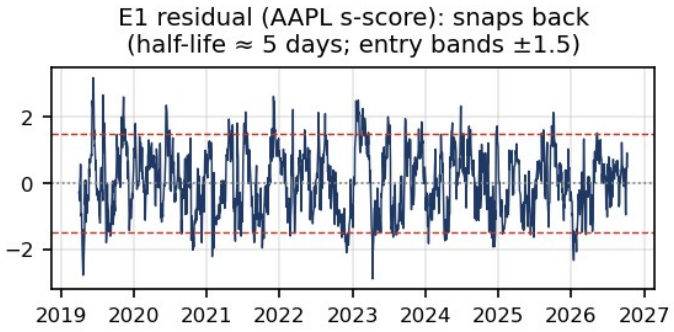
Implied and GARCH agree as forecasts; both sit above what volatility delivered. The ≈5-vol-point average gap is the implied-minus-realised vol spread — the premium, quoted in vol points. The wheel sells that gap and nets assignment losses against it in the backtest (Sleeve 2); the chart's 30-day IV input flatters calm-market levels, as disclosed.

Cointegration exhibits: beta without an anchor

Pair	β	EG p	Spread HL
ADRO vs coal (BTU)	0.39	0.95	>500d
PTBA vs coal (BTU)	0.24	0.83	312d
ITMG vs coal (BTU)	0.49	0.79	325d
BYAN vs coal (BTU)	0.55	0.73	>500d
ANTM vs gold	1.40	0.19	153d
MDKA vs gold	1.47	0.89	497d

Daily log prices, Jan 2016 to Oct 2026. β from OLS of log miner on log commodity; EG p is the Engle–Granger cointegration p-value; HL is the spread's AR(1) half-life.





Engle–Granger verdict: no pair cointegrates (all p ≥ 0.19; spread half-lives of 5 months to effectively never). The miners carry commodity *return beta* without a price-level anchor, so Sleeve 3's design research used trend gates, not spreads against the commodity.

Contrast, Sleeve 1: ADF rejects a unit root for **all 34** factor-residual series; residual half-lives cluster at 5 to 6 days, inside the engine's 30-day admission filter. That is what a tradeable spread looks like.

Sleeve 4's filter is the same logic used defensively: dip entries require a close above the 200-day average, so in 2022 the sleeve returned −1.1% while sitting mostly in cash (average capital use 10%). An uptrend filter is a crude but effective regime gate.